Empirical Methods: Assignment 5

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Data: `education_income.csv`
Total points: 5 points, with 1 point per question.
Long Answers: For each question, and if not otherwise specified, keep your answer under 150 words (0.5 points per question are removed if the answer is longer).

In this assignment, we examine the relationship between education and income. The aim is to understand treatment effect heterogeneity, functional form assumptions, and the identification of causal effects using individual-level data. We study a cohort of high school graduates who applied to a government college scholarship program. The program first screened applicants according to eligibility criteria. Students who passed this initial selection stage entered the scholarship program, and the amount of scholarship support they received was then assigned by the program. The program was run twice, once in 2000 and once in 2010. The dataset contains information on students' college attendance, scholarship amount, household income, race group, and income later in life. The main treatment of interest is college attendance. The main outcome of interest is individual income, measured 10 years after graduation. The key causal question is whether attending college increases long-run income.

Dataset

The dataset contains the following variables:

Variable	Description
`college`	1 = attended college, 0 = did not
`income`	Individual income in 1000 USD
`scholarship`	Scholarship amount received
`hhincome`	Household income in 1000 USD
`race`	Race group indicator, equal to 1 or another value
`year`	Year of observation, 2000 or 2010

Please use the same data cleaning procedure used in Exercise 5!

Question 1: Descriptive Analysis [0.25 point]

Load the dataset and compute basic descriptive statistics.

(a) Compute mean income for college attendees and non-attendees. Can we interpret raw mean income difference between college and non-college individuals as the causal effect of attending college on income? Why or why not?

Question 2: Assignment Mechanism [2 point]

A researcher estimates the OLS regression

$$income_i = \beta_0 + \beta_1 college_i + u_i$$

and obtains $\hat{\beta}_1 > 0$.

(a) Using the potential outcomes notation from class, decompose the observed mean difference

$$E(Y_i \mid D_i = 1) - E(Y_i \mid D_i = 0)$$

into the Average Treatment Effect on the Treated (ATET) and a selection bias term. Write the equation.

(b) Do you think that in this setting $\hat{\beta}_1$ measures the causal effect of interest or is it a biased estimate? Briefly explain your answer.

(c) If you think that it is a biased estimate, which variable in the dataset could be a key driver of selection into college? If you think the estimate is correct, skip the following questions.

(d) Estimate the regression in (a) and then add observed controls for household income, race, and year. Compare the estimated coefficient on `college` across the two regressions. Does adding these controls address the selection-on-observables concern? Explain briefly.

(e) Now consider a setting in which scholarship amounts are randomly assigned to students. In this hypothetical experiment, explain why random assignment would eliminate the selection bias term from part (a).

(f) Estimate a two-stage least squares model using `scholarship` as an instrument for `college`, controlling for household income, race, and year.

Comment the first stage regression. Then report the IV estimate of the effect of `college` on income and compare it to the OLS estimate with controls. In one or two sentences, discuss whether the IV estimate should be interpreted as causal in this setting.

Question 3: Effect Heterogeneity [1 point]

We now ask whether the income return to college differs across race groups.

(a) Write the regression equation for income that includes college, race, and their interaction. Which coefficient captures the heterogeneous effect, and how is it interpreted?

(b) Estimate the equation from (a) in R.

(c) Using the estimated coefficients, report the marginal effect of college on income separately for `white == 1` and `white != 1`.

(d) Is the difference in the college income premium across race groups statistically significant? What does this tell us about effect heterogeneity in the return to education?

Question 4: Probit and Logit Models [1 point]

We model the probability of attending college, $P(college = 1)$, as a function of household income and race.

(a) State one specific limitation of using OLS, when the dependent variable is binary.

(b) Estimate a probit and a logit model in R using household income and race as covariates. Comment on the sign, magnitude, and significance of each coefficient. Do both models agree on the direction of the effects?

Question 5: Marginal Effects [0.75 point]

Raw probit and logit coefficients do not represent marginal effects on the probability scale.

(a) Write the formula for the Average Partial Effect (APE) of a continuous variable x_j in a probit model.

(b) Compute and interpret the APE of `hhincome` from the probit model estimated in 4(b).

(c) Compute the predicted probability of attending college for an individual with `hhincome == 80` and `white == 1`. Verify that the predicted value lies between 0 and 1.